Clostridium difficile colitis in the United States: a decade of trends, outcomes, risk factors for colectomy, and mortality after colectomy.

BACKGROUND: Clostridium difficile colitis (CDC) is a major health concern in the United States (US), with earlier reports demonstrating a rising incidence. Studies analyzing predictors for total colectomy and mortality after colectomy are limited by small numbers. STUDY DESIGN: The Nationwide Inpatient Sample (NIS) 2001 to 2010 was retrospectively reviewed for CDC trends, the associated colectomy and mortality rates. Patient and hospital variables were used in the LASSO algorithm for logistic regression with 10-fold cross validation to build a predictive model for colectomy requirement and mortality after colectomy. The association of colectomy day with mortality was also examined on multivariable logistic regression analysis. RESULTS: An estimated 2,773,521 discharges with a diagnosis of CDC were identified in the US over a decade. Colectomy was required in 19,374 cases (0.7%), with an associated mortality of 30.7%. Compared with the 2001 to 2005 period, the 2006 to 2010 period witnessed a 47% increase in the rate of CDC and a 32% increase in the rate of colectomies. The LASSO algorithm identified the following predictors for colectomy: coagulopathy (odds ratio [OR] 2.71), weight loss (OR 2.25), teaching hospitals (OR 1.37), fluid or electrolyte disorders (OR 1.31), and large hospitals (OR 1.18). The predictors of mortality after colectomy were: coagulopathy (OR 2.38), age greater than 60 years (OR 1.97), acute renal failure (OR 1.67), respiratory failure (OR 1.61), sepsis (OR 1.40), peripheral vascular disease (OR 1.39), and congestive heart failure (OR 1.25). Surgery more than 3 days after admission was associated with higher mortality rates (OR 1.09; 95% CI 1.05 to 1.14; $p < 0.05$). CONCLUSIONS: Clostridium difficile colitis is increasing in the US, with an associated increase in total colectomies. Mortality rates after colectomy remain elevated. Progression to colectomy and mortality thereafter are associated with several patient and hospital factors. Knowledge of these risk factors may help in risk-stratification and counseling.